Polytropic Star

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Abstract

This note derives the structure and global properties of a polytropic star, $P = K\rho^{1+1/n}$. We first solve the Lane-Emden equation and express the central density ρ_c, the entropy constant K, and the central pressure in terms of the stellar mass M and radius R, which yields the mass-radius relation $R \propto M^{(n-1)/(n-3)}$ at fixed K. We then obtain the moment of inertia $I = kMR^2$, the central potential $\Phi_c = -(1 + \xi_{\rm max}/\phi_{\rm max})GM/R$, and the gravitational energy $\Omega = -3GM^2/[(5-n)R]$, together with polynomial fits accurate to $< 0.2\%$ for $0.5 < n < 3$. Combined with the virial theorem, these results show that a star is bound only if its adiabatic index exceeds $4/3$. Next, we show that a gas-pressure-dominated radiative envelope with Kramers' opacity follows $P \propto \rho^{17/13}$ ($n = 13/4$), close to an $n = 3$ polytrope, whereas a radiation-pressure-dominated envelope must radiate at the Eddington luminosity. Finally, for the negligible-mass surface layers ($r \approx R$), we derive analytic density, pressure, and exterior-mass profiles and show that $P \approx [GM^2/(4\pi R^4)]\, M_{\rm ex}/M$.

1. Lane-Emden Equation

Defining $\rho = \rho_c\theta^n$, $\gamma = 1 + 1/n$, $P = K\rho^\gamma = K\rho_c^{1+1/n}\theta^{n+1}$, $\alpha^2 = (n+1)K\rho_c^{1/n-1}/(4\pi G)$, and $\xi = r/\alpha$, the equation of hydrostatic equilibrium ($\mathrm{d}P/\mathrm{d}r = -G\rho m(r)/r^2$) and mass continuity ($\mathrm{d}m/\mathrm{d}r = 4\pi\rho r^2$) can be written in the following Lane-Emden form

$$\frac{1}{\xi^2}\frac{\mathrm{d}}{\mathrm{d}\xi}\left(\xi^2\frac{\mathrm{d}\theta}{\mathrm{d}\xi}\right) = -\theta^n. \tag{1}$$

The inner boundary conditions are $\theta(\xi = 0) = 1$ and $\mathrm{d}\theta/\mathrm{d}\xi(\xi = 0) = 0$ (since $\mathrm{d}P/\mathrm{d}r = 0$ at $r = 0$).

The above equation can be solved by defining a vector $\vec{X} = (\theta, \phi)^{\rm T}$, where ϕ is defined as

$$\phi \equiv -\xi^2\mathrm{d}\theta/\mathrm{d}\xi, \;\; \mathrm{d}\phi/\mathrm{d}\xi = \xi^2\theta^n. \tag{2}$$

The spatial gradient of $\vec{X}$ is given by

$$\frac{\mathrm{d}\vec{X}}{\mathrm{d}\xi} = (-\phi/\xi^2, \xi^2\theta^n)^{\rm T}, \tag{3}$$

and the inner boundary condition is $\vec{X}(\xi = 0) = (1, 0)$. We integrate the above vector equation from $\xi = 0$ to a maximum $\xi_{\rm max}$ where θ has decreased to 0. It can be seen that $\phi > 0$ is a monotonically increasing function of ξ and $\theta(\xi)$ is a monotonically decreasing function. The solutions for $n = 1.5$ ($\gamma = 5/3$) and $n = 3$ ($\gamma = 4/3$) are shown in Fig. 1.

A physical solution requires θ to reach zero at a finite $\xi_{\rm max}$ (i.e., a finite stellar radius R). This is the case only for $0 \le n < 5$ (cf. Chandrasekhar). For $n < 0$, the density $\rho = \rho_c\theta^n$ would increase outward as θ decreases. For $n = 5$, the analytic solution $\theta = (1 + \xi^2/3)^{-1/2}$ never reaches zero,

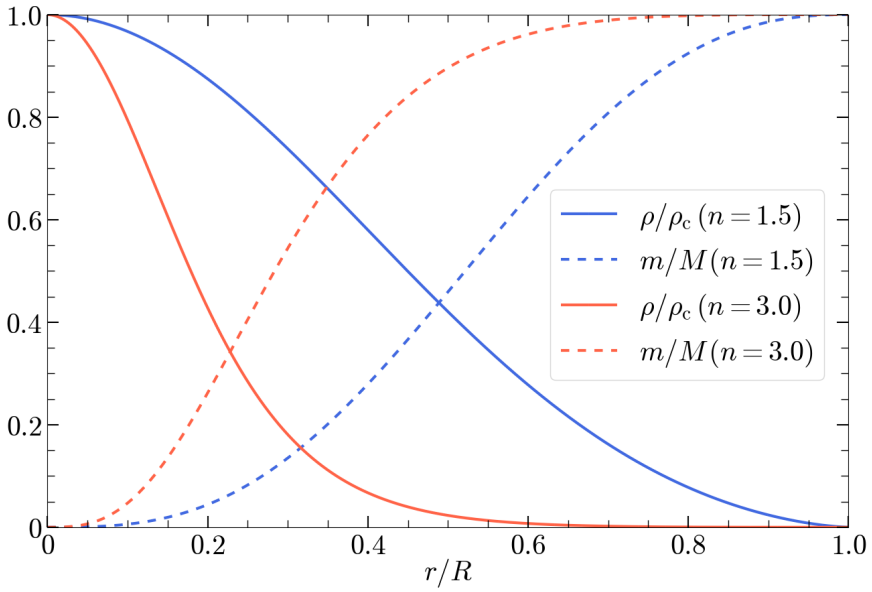


Fig. 1.— Polytropic density $\rho(r)$ and enclosed-mass $m(r)$ profiles, which are normalized by the central density ρ_c and the total mass M, respectively. The radius r is normalized by the total radius R. Note that $\rho/\rho_c = \theta^n$, $r/R = \xi/\xi_{\rm max}$, and $m/M = \phi/\phi_{\rm max}$.

so $R \to \infty$ even though the total mass is finite; for $n > 5$, the mass is also infinite. We restrict ourselves to $0 \le n < 5$ in the rest of this note.

It should be noted that since $d\phi/d\xi = \xi^2\theta^n \propto r^2\rho \propto dm/dr$, $\phi(\xi)$ is proportional to the mass coordinate:

$$\phi(\xi) = \frac{m(r)}{4\pi\alpha^3\rho_c}. \tag{4}$$

Once the functions $\theta(\xi)$ and $\phi(\xi)$ have been numerically obtained (including the value of $\xi_{\rm max}$), we can solve for the two constants ρ_c and K from the physical stellar mass M and radius R by

$$\phi_{\rm max} \equiv \phi(\xi_{\rm max}) = \frac{M}{4\pi\alpha^3\rho_c},\ \alpha\xi_{\rm max} = R. \tag{5}$$

The results are

$$\rho_c = \frac{M/\phi_{\rm max}}{4\pi(R/\xi_{\rm max})^3},\ K = \frac{G}{n+1}(4\pi)^{1/n}(M/\phi_{\rm max})^{1-1/n}(R/\xi_{\rm max})^{3/n-1}. \tag{6}$$

The central pressure is then given by

$$P_c = \frac{1}{n+1}\frac{G(M/\phi_{\rm max})^2}{4\pi(R/\xi_{\rm max})^4}. \tag{7}$$

For a constant K, we obtain a mass-radius relation

$$R \propto M^{(n-1)/(n-3)},\quad \text{for } K = \text{const.} \tag{8}$$

Such a mass-radius relation applies to low-mass white dwarfs that are supported by non-relativistic degeneracy pressure as K is related to fundamental constants.

Another interesting situation is that, when a fully convective star (where K is uniform and $\gamma = 5/3$ or $n = 1.5$) undergoes mass loss on a timescale shorter than the thermal timescale of its outer envelope, we expect K to remain constant despite the adjustment of the remaining star to reach hydrostatic equilibrium (as K only evolves on a thermal timescale) and the mass-radius relation predicts $R \propto M^{-1/3}$. We find that a fully convective star expands as it loses mass rapidly on timescales shorter than the thermal timescale but longer than the dynamical timescale.

The moment of inertia of a spherical shell of thickness dr is d$I = (8\pi/3)\rho(r)r^4$dr, which can be obtained by differentiating $I = (2/5)MR^2$ for a uniform sphere. Then, the moment of inertia of the entire star is given by

$$I = \frac{8\pi}{3}\rho_{\rm c}\alpha^5\tilde{I} = \frac{2}{3}\frac{MR^2}{\phi_{\rm max}\xi_{\rm max}^2}\tilde{I}, \tag{9}$$

where

$$\tilde{I} \equiv \int_0^{\xi_{\rm max}} \theta^n \xi^4 {\rm d}\xi. \tag{10}$$

It is common to define a convenient constant

$$k = (2/3)\tilde{I}/(\phi_{\rm max}\xi_{\rm max}^2), \tag{11}$$

such that $I = kMR^2$.

The central potential of the star is given by the sum of the contribution from all spherical shells[1]

$$\Phi_{\rm c} = -G\int_0^R \frac{{\rm d}m}{r} = -4\pi\alpha^2 G\rho_{\rm c}\int_0^{\xi_{\rm max}}\frac{{\rm d}\phi}{\xi} = -\frac{GM}{R}\frac{\xi_{\rm max}}{\phi_{\rm max}}\tilde{\Phi}_{\rm c}, \tag{12}$$

where we have used ${\rm d}\phi/{\rm d}\xi = \xi^2\theta^n$, $M/R = 4\pi\alpha^2\rho_{\rm c}\phi_{\rm max}/\xi_{\rm max}$, and

$$\tilde{\Phi}_{\rm c} = \int_0^{\xi_{\rm max}}\theta^n\xi{\rm d}\xi = \int_0^{\xi_{\rm max}}\xi^{-1}({\rm d}\phi/{\rm d}\xi){\rm d}\xi = \int_0^{\xi_{\rm max}}\xi^{-1}{\rm d}\phi, \tag{13}$$

where we have used ${\rm d}\phi/{\rm d}\xi = \xi^2\theta^n$. Integrating the last expression above by parts, we obtain

$$\tilde{\Phi}_{\rm c} = (\phi/\xi)|_{\xi=0}^{\xi=\xi_{\rm max}} + \int_0^{\xi_{\rm max}}\phi\xi^{-2}{\rm d}\xi = \frac{\phi_{\rm max}}{\xi_{\rm max}} - \int_0^{\xi_{\rm max}}({\rm d}\theta/{\rm d}\xi){\rm d}\xi = \frac{\phi_{\rm max}}{\xi_{\rm max}} + 1, \tag{14}$$

where we have used $\phi = -\xi^2{\rm d}\theta/{\rm d}\xi$ and boundary conditions: $\theta(\xi = 0) = 1$, $\theta(\xi_{\rm max}) = 0$, and $\phi/\xi \approx \xi^2/3 \to 0$ as $\xi \to 0$. We find that the central potential of a polytropic star is given by

$$\Phi_{\rm c} = -\frac{GM}{R}\left(1 + \frac{\xi_{\rm max}}{\phi_{\rm max}}\right). \tag{15}$$

[1]Remember that the potential inside a spherical shell of mass dm and radius r is constant and equal to the surface value of $-G{\rm d}m/r$.

Polynomial fits to the numerical results are

$$\begin{aligned}
\xi_{\max} &\approx 0.05583n^4 - 0.2165n^3 + 0.5656n^2 + 0.1914n + 2.544, \\
\phi_{\max} &\approx 0.02393n^4 - 0.2363n^3 + 0.989n^2 - 2.402n + 4.767, \\
k = I/(MR^2) &\approx 2.112 \times 10^{-4}n^4 - 1.969 \times 10^{-3}n^3 + 0.02006n^2 - 0.1561n + 0.3992,
\end{aligned} \tag{16}$$

which are accurate to fractional errors $< 0.2\%$ for $n \in (0.5, 3)$ (the above k fit has maximum fractional error of 3.5×10^{-4}).

The gravitational potential energy of the star is given by

$$\Omega = -G \int_0^M \frac{m(r)\,\mathrm{d}m}{r} = -\frac{GM^2}{R}\frac{\xi_{\max}}{\phi_{\max}^2}\tilde{\Omega}, \tag{17}$$

where

$$\tilde{\Omega} = \int_0^{\xi_{\max}} \phi(\xi)\theta^n \xi \mathrm{d}\xi = -\int_0^{\xi_{\max}} \theta^n \xi^3 \frac{\mathrm{d}\theta}{\mathrm{d}\xi}\mathrm{d}\xi = -\frac{1}{n+1}\int_0^{\xi_{\max}} \xi^3 \mathrm{d}\theta^{n+1}. \tag{18}$$

In the above derivation, we have used $\phi = -\xi^2 \mathrm{d}\theta/\mathrm{d}\xi$. Integrating the final expression above by parts and making use of $\theta(\xi_{\max}) = 0$, we find

$$\tilde{\Omega} = \frac{3}{n+1}\int_0^{\xi_{\max}} \theta^{n+1}\xi^2 \mathrm{d}\xi. \tag{19}$$

We skip the details here (cf. Chandrasekhar) but it is possible to manipulate the Lane-Emden equation to prove the following identity

$$\int_0^{\xi_{\max}} \theta^{n+1}\xi^2 \mathrm{d}\xi = \frac{n+1}{5-n}\frac{\phi_{\max}^2}{\xi_{\max}}, \tag{20}$$

so the potential energy is simply given by

$$\Omega = -\frac{3}{5-n}\frac{GM^2}{R}. \tag{21}$$

Eq. (21) is valid only for $0 \leq n < 5$, where $\xi_{\max}$ is finite; the divergence of Ω as $n \to 5$ reflects $\xi_{\max} \to \infty$. Based on the equation of hydrostatic equilibrium, we also know that the potential energy is connected to the volume integral of the pressure via (so-called virial theorem)

$$\Omega = -3\int P\,\mathrm{d}V, \tag{22}$$

and the RHS of the above equation is related to the internal energy of the star.

Let u denote the internal energy density such that the total internal energy is

$$U = \int u\,\mathrm{d}V. \tag{23}$$

Consider the adiabatic evolution of a fluid element, the first law of thermodynamics tells us

$$\mathrm{d}\left(\frac{u}{\rho}\right) = -P\,\mathrm{d}\left(\frac{1}{\rho}\right). \tag{24}$$

If we *assume* a constant adiabatic index $\gamma_{\rm ad} \equiv (\mathrm{d}\ln P/\mathrm{d}\ln\rho)_{\rm ad} = \text{const}$ (although this is not generally true), then integrating along an adiabat $P \propto \rho^{\gamma_{\rm ad}}$ gives

$$u = \frac{P}{\gamma_{\rm ad}-1}, \text{ if } \gamma_{\rm ad} = \text{const.} \tag{25}$$

In this case, the total internal energy is given by

$$U = \frac{1}{\gamma_{\rm ad}-1}\int P\,\mathrm{d}V = -\frac{\Omega}{3(\gamma_{\rm ad}-1)} = \frac{1}{(\gamma_{\rm ad}-1)(5-n)}\frac{GM^2}{R}, \text{ if } \gamma_{\rm ad} = \text{const.} \tag{26}$$

Finally, the binding energy of a star is given by

$$E_{\rm tot} = U + \Omega = -\frac{3\gamma_{\rm ad}-4}{(\gamma_{\rm ad}-1)(5-n)}\frac{GM^2}{R} \tag{27}$$

For a realistic, bound star, we require $E_{\rm tot} < 0$ and hence

$$\gamma_{\rm ad} > 4/3. \tag{28}$$

Note that the sign of the total energy is controlled by $\gamma_{\rm ad}$ but not by the polytropic index n (which tells how the star is radially stratified). In fact, $\gamma_{\rm ad}$ determines how a fluid element responds to adiabatic compression and therefore enters the internal energy and the dynamical stability.

In the special case where the polytropic stratification coincides with an adiabat, i.e., the entire star has isentropic structure and a constant $\gamma_{\rm ad}$ (as is the case for a fully convective star made of classical ideal gas or a star supported by non-relativistic electron degeneracy pressure), we have $\gamma_{\rm ad} = \gamma = 1 + 1/n$ (and also $\nabla = \nabla_{\rm ad}$), and we find the total binding energy to be

$$E_{\rm tot} = -\frac{3-n}{5-n}\frac{GM^2}{R}, \text{ if } \gamma_{\rm ad} = \gamma = \text{const.} \tag{29}$$

2. Radiative Envelope where $m(r) \approx M$ and $L(r) \approx L$

The polytropic stratification works well for the convective envelope of a star where the entropy is constant and we have $P \propto \rho^{5/3}$.

However, for a radiative envelope, one needs to consider the radiative diffusion

$$\frac{L}{4\pi r^2} = -\frac{c}{3\rho\kappa}\frac{\mathrm{d}(aT^4)}{\mathrm{d}r}. \tag{30}$$

This should be combined with the equation for hydrostatic equilibrium

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\rho\frac{GM}{r^2}. \tag{31}$$

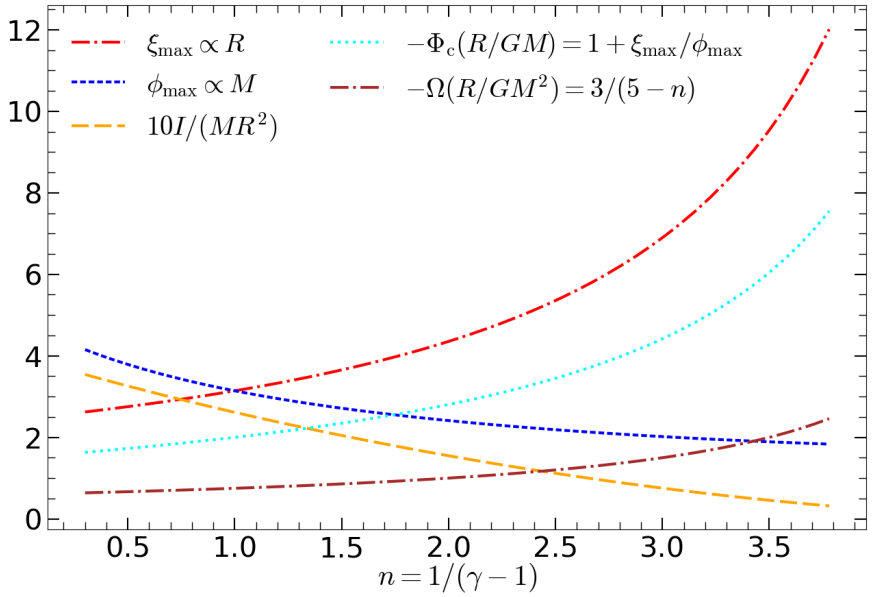


Fig. 2.— The integral quantities $\xi_{\rm max}$ (related to stellar radius), $\phi_{\rm max}$ (related to stellar mass), $k \equiv I/(MR^2)$ (related to the moment of inertia), $\Phi_{\rm c}$ (central potential), and Ω (gravitational potential energy) for a polytropic star. Here, we show $10k$ (instead of k) for clarity.

Taking the ratio between these two equations, we obtain

$$\frac{T^3}{\kappa}\frac{{\rm d}T}{{\rm d}P} = \frac{3L}{16\pi acGM}. \tag{32}$$

Since we are considering the outer envelope, the right-hand side of the above equation is nearly a constant.

Let us then consider a particular form of Rosseland-mean opacity $\kappa = \kappa_0 P^q T^{-q-s}$. For Kramers' law under gas pressure, one has $q = 1$ and $s = 3.5$. Then, the relation between T and P is given by

$${\rm d}T^{4+q+s} = C\,{\rm d}P^{q+1},\ C = \frac{4+q+s}{q+1}\frac{3\kappa_0 L}{16\pi acGM}. \tag{33}$$

One can integrate the above equation from the outer boundary (near the photosphere) where $T = T_{\rm ph}$ and $P = P_{\rm ph}$. Then, in the regions far below the photosphere where $T \gg T_{\rm ph}$ and $P \gg P_{\rm ph}$, one obtains the following

$$T^{4+q+s} = C\,P^{q+1}, \tag{34}$$

which means (for the gas pressure dominated case)

$$P \propto \rho^{1+(q+1)/(s+3)}. \tag{35}$$

For Kramers' law ($q = 1$ and $s = 3.5$), we have $P \propto \rho^{17/13 \approx 1.308}$ or $n = 13/4 \approx 3.25$, which is very close to the case of a $\gamma = 4/3$ (or $n = 3$) polytrope. The proportional constant $K = P/\rho^\gamma$ depends on the stellar luminosity and mass. This applies to stars on the upper main-sequence with masses greater than about $1M_\odot$ (for lower-mass stars, there is a substantial convective envelope due to higher opacity in the outer layers).

In the extreme limit where the pressure is dominated by radiation $P \simeq aT^4/3$ (although this is usually not fully realized), then the radiative transfer equation can be written as

$$\rho \frac{L}{4\pi r^2} = -\frac{c}{\kappa}\frac{\mathrm{d}P}{\mathrm{d}r}. \tag{36}$$

Taking the ratio between the above equation and the hydrostatic equilibrium equation, one obtains

$$L = \frac{4\pi GMc}{\kappa} = L_{\rm Edd}. \tag{37}$$

In order to maintain locally Eddington luminosity everywhere, the opacity must be nearly constant over the envelope. This can be achieved if κ is dominated by electron scattering.

3. Structure of the surface layers where $r \approx R$

The structure of the outermost layers of the star is very simple when we ignore the self-gravity. From hydrostatic equilibrium and the equation of state, we obtain

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{GM\rho}{r^2}, P = K\rho^\gamma \;\Rightarrow\; \mathrm{d}\rho^{\gamma-1} = \frac{(\gamma-1)GM}{\gamma K}\mathrm{d}r^{-1}. \tag{38}$$

This can be easily integrated using the boundary condition of $\rho(r = R) = 0$,

$$\rho^{\gamma-1}(r) = \frac{(\gamma-1)GM}{\gamma K}\left(\frac{1}{r}-\frac{1}{R}\right), \text{ or } \rho(r) = \left[\frac{GM}{(n+1)K}\left(\frac{1}{r}-\frac{1}{R}\right)\right]^n. \tag{39}$$

The pressure profile is given by

$$P(r) = K\left[\frac{GM}{(n+1)K}\left(\frac{1}{r}-\frac{1}{R}\right)\right]^{n+1}. \tag{40}$$

Using K in eq. (6), we obtain

$$P(r) = \frac{GM^2}{4\pi R^4}\frac{(R/r-1)^{n+1}}{n+1}\phi_{\rm max}^{n-1}\xi_{\rm max}^{3-n}. \tag{41}$$

This means that the pressure scaleheight near the surface is of the order $H \sim R - r$ (for $r \approx R$). Another consequence is that the isothermal sound speed $c_{\rm s} \equiv \sqrt{P/\rho}$ has the following simple form

$$c_{\rm s}^2 = K\rho^{\gamma-1} = \frac{GM\,(R-r)}{(n+1)rR}. \tag{42}$$

The exterior mass is given by

$$\begin{aligned} M_{\rm ex}(r) &= \int_r^R 4\pi(r')^2\rho(r')\mathrm{d}r' \\ &= 4\pi\left(\frac{(\gamma-1)GM}{\gamma K}\right)^{\frac{1}{\gamma-1}} R^{\frac{3\gamma-4}{\gamma-1}}\int_{r/R}^1 x^{\frac{2\gamma-3}{\gamma-1}}(1-x)^{\frac{1}{\gamma-1}}\mathrm{d}x, \end{aligned} \tag{43}$$

or, using $\gamma = 1 + 1/n$,

$$M_{\rm ex}(r) = 4\pi \left(\frac{GM}{(n+1)K}\right)^n R^{3-n} \int_{r/R}^{1} x^{2-n}(1-x)^n \, {\rm d}x. \tag{44}$$

Using the entropy constant K in eq. (6), we obtain

$$\frac{M_{\rm ex}(r)}{M} = \phi_{\rm max}^{n-1} \xi_{\rm max}^{3-n} \int_{r/R}^{1} x^{2-n}(1-x)^n \, {\rm d}x. \tag{45}$$

In the limit $r/R \approx 1$, this can be further simplified into

$$\frac{M_{\rm ex}(r)}{M} \approx \phi_{\rm max}^{n-1} \xi_{\rm max}^{3-n} \frac{(1-r/R)^{n+1}}{n+1}, \text{ for } 1 - r/R \ll 1. \tag{46}$$

In the same limit, the pressure profile is given by

$$\frac{P(r)}{\bar{P}} \approx \phi_{\rm max}^{n-1} \xi_{\rm max}^{3-n} \frac{(1-r/R)^{n+1}}{n+1}, \text{ for } 1 - r/R \ll 1, \tag{47}$$

where $\bar{P} \equiv GM^2/(4\pi R^4)$ is a rough estimate of the mean pressure inside the star. Thus, we have arrived at the following interesting result

$$\frac{P(r)}{\bar{P}} \approx \frac{M_{\rm ex}(P)}{M}, \tag{48}$$

where $M_{\rm ex}(P)$ is the mass exterior to a critical radius specified by a given pressure P.